User Defined Protocol: protocol driver

Overview

The UserDrv protocol driver for Adroit is a generic paging and text device driver. Messages configured in String or Notifier Agents may be sent via an encoder or modem to paging devices. Alarms that occur in Adroit can thus be routed to automatically page a person.

This driver is intended for easy use with many varied text protocols, while allowing advanced users to configure custom protocols of their own. No windows drivers for a modem (is used) are needed, this driver provides it's own interface and control over the modem via the serial port or IP address assigned to it.

Sections

User Defined Protocol: protocol driver....1
Overview....1
Driver Installation and Setup....1
Wiring....7
Supported Addresses....7
Agent, SMS alarm and Scanning Configuration....8
Data Acquisition Devices....10
Cellular GSM Modems....13
Custom Protocols....15
Message protocols....22
Modem reference....24
Troubleshooting....25
Abbreviations....28
Revision information....28

Driver Installation and Setup

Installation of the UserDrv protocol driver is done in the standard way in Adroit Set-up. After installation of the driver, the user may then create and configure devices for the UserDrv driver.

Devices

At a minimum, you must create at least one device for each encoder/modem. Currently, the port settings for all devices on a port must be identical. I.e.: It is not permitted for example to use Vodacom and AutoPage protocols together if they use different data bit settings.

Device configuration dialog

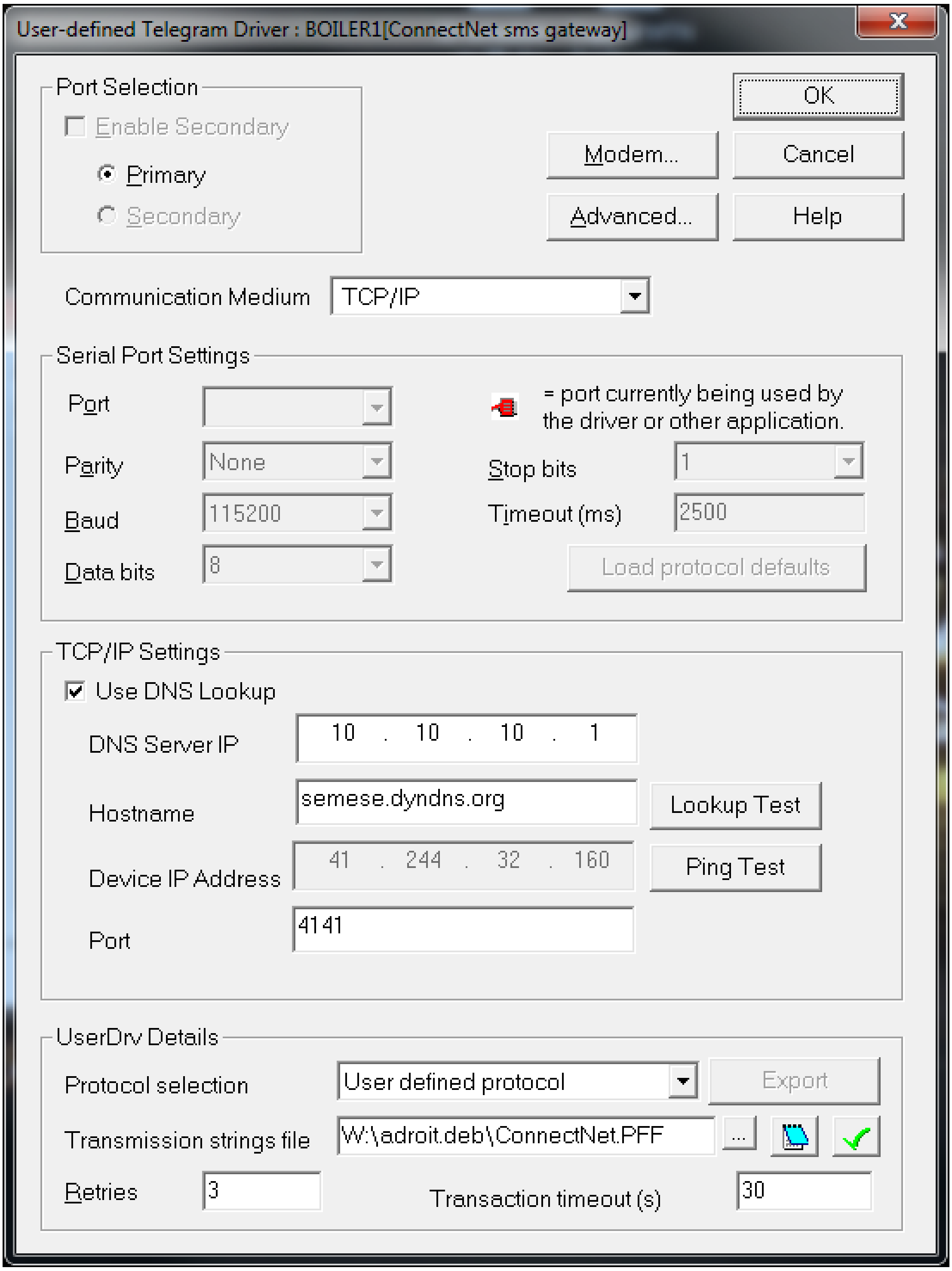


Fig. 1

UserDrv Details settings

Communication medium Used to select whether TCP/IP or serial will be used as the communications medium.

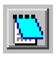
Port	The communication port to which the modem/device is attached. A coloured icon helps indicate if the port is used or not, (red icon replaces the asterisk that used to appear).
Parity	The type of parity, odd, even or none.
Baud	The baud-rate, in the case of a modem, select the highest usable speed.
Data bits	The number of data bits will depend on the service used.
Stop Bits	The number of stop bits to use will depend on the service used.
Timeout	The time to wait for a response from the locally connected device.
Use DNS Lookup	Checking this box will use the DNS Server IP and hostname to do a lookup of the Device IP Address. This allows devices to be referenced which may have dynamic IP Addresses.
DNS Server IP	The IP address of the server to use for DNS lookup.
Hostname	The hostname of the device for which the IP address is to be requested from the DNS server.
Lookup Test	Clicking this button does a DNS lookup and brings up a dialog box showing the IP address retrieved.
Device IP Address	The IP address that will be used to communicate with the device. If the “Use DNS Lookup” box is checked this IP address will be the result of a DNS lookup based on the DNS Server IP and Hostname
Ping Test	Clicking this button will do a ping test and display the result in a dialog box
Port	This is the port number that adroit will use when communicating with the device

Protocol selection	Select a pre-defined protocol or “User defined” if you want to use a definition file. When selecting a predefined protocol, the next field is ignored.
Transmission strings file	The file to use if selecting a definition file, selection is disabled and ignored if “User defined” is not selected.
Retries	Number of attempts to communicate.
Transaction timeout (s)	Total number of seconds before a transaction will be deemed as impossible. This setting is a safety stop in case the driver gets stuck or the remote device keeps giving the same bad response. (This figure is added to the time used to dial up if using a modem)

Setting screen notes:

- If using AutoPage, select 7 data, no parity with one stop bit.
- If using Vodacom, MTN or Zetron , select 8 data, no parity, one stop bit.
- The Communications Script is called a **P**rotocol **F**ormat **F**ile or .PFF file.

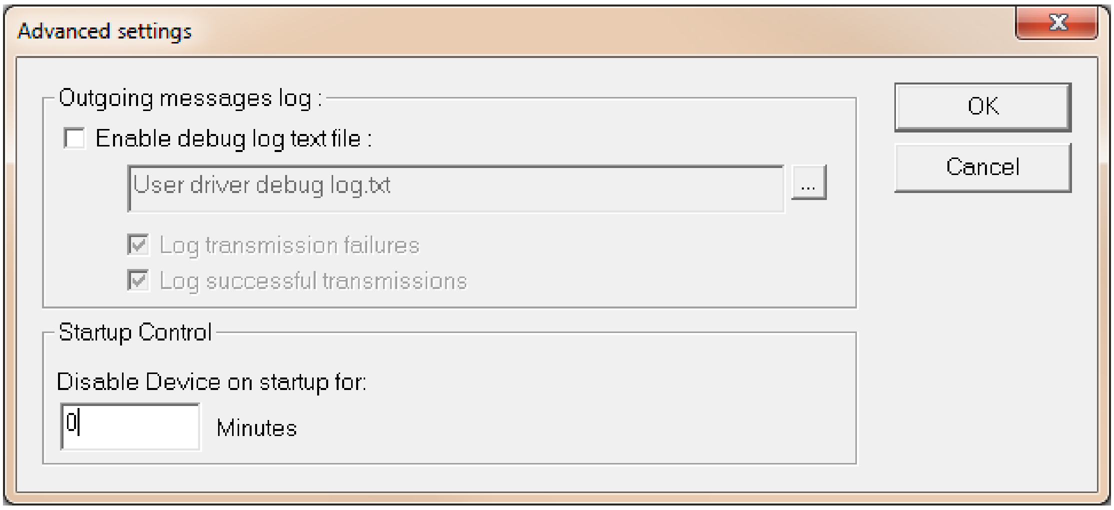
- Clicking “Load protocol defaults” will apply certain relevant configuration settings from the script to this device’s settings.



- The “Notepad” button will allow you to edit a script if it is a custom script, or if it has been exported.

- The “check” button opens a dialog showing the parsed script. If there are any errors, they will be shown. REMEMBER, when using a built-in protocol, the PFF file if present is ignored at runtime.

Advanced configuration Dialog



Enable Debug log text file	Enables logging of debugging data to a specified text file.
Log transmission failures	Enables logging of transmission failures to the text file.
Log successfull transmissions	Enables/Disables logging of successful transmissions to the text file
Disable device on startup for	This will inhibit the sending of write messages for a user configurable period after the device is started. This is defaulted to zero. **Use with extreme caution as this can very well cause alarm notifications to not be sent.**

Modem configuration dialog

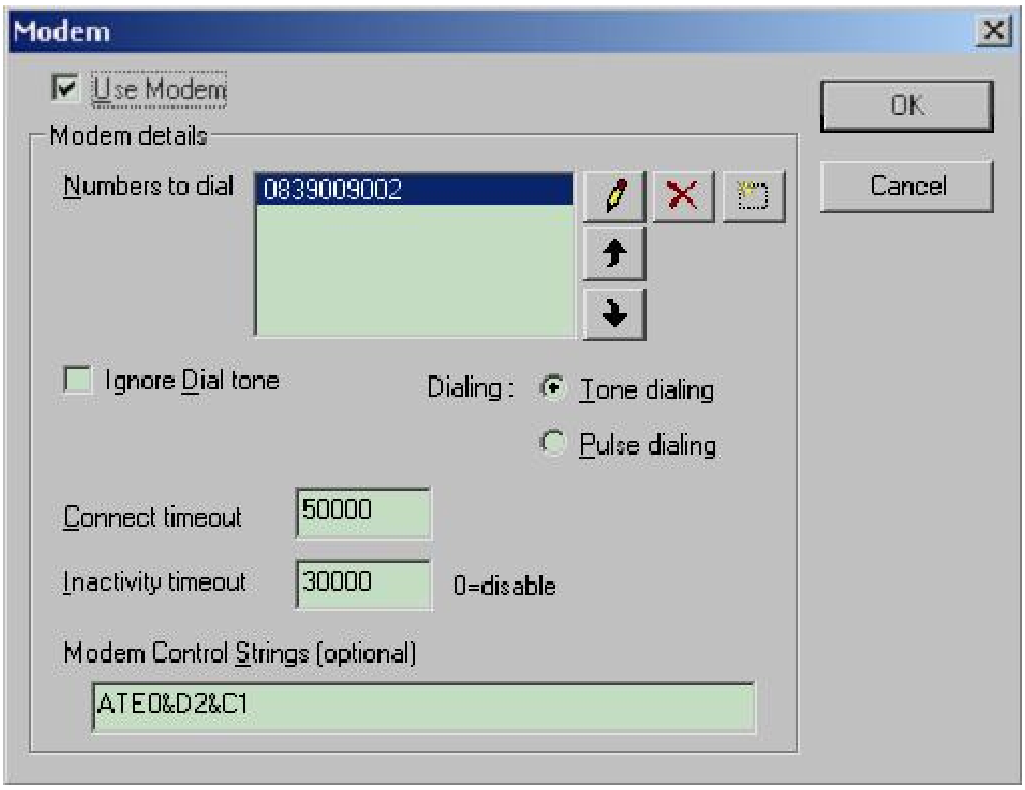


Fig. 2

The Modem button (See Fig. 1) will bring up the Modem Settings dialogue (Fig. 2 above).

Use modem - Will enable modem dialling and carrier detection features here. **If you do not use a modem, clear this tick-mark.**

Ignore Dial tone - Must to be ticked if a dial tone is not always present or very clear. If line detection problems persist, try putting extra commas in before the number to dial. Turn this box on if going via a PABX were you need to put a “0,” in front of the number.

Numbers to dial - The telephone number, which must be dialled to create a connection. (More detail below)

Tone / Pulse – Tone dialling may not be available in your area, if in doubt use pulse dialling which is a bit slower.

Connect timeout – The total amount of time from the commencement of dialling, until the modems finish connecting. (Carrier detected), recommended value 45000 is 45 seconds.

Inactivity timeout – The amount of time the driver will “keep” an idle connection open for in order to allow sending of additional messages without having to redial (must be supported by the protocol however). Recommended value of 30000 is 30 seconds. Use 0 if you want the modem to hang up immediately after a transaction completes.

Modem control String – The default modem control string is ATE0&D2&C1. If no string is specified, the driver will use the default string.

Modem setting screen notes:

- For an **Auto-Page** dial-up you need to disable compression and error correction features; on newer modems, use the following string. ATE0&D2S15=255&C1&W0. This will turn off ARQ, MNP features on the US robotics modems.

- The **Phone number entry** may need a “0,” before the number to allow for dialling out via switchboard, in which case you must turn on the “ignore DT” setting. Try using additional commas after the “0” or even before the “0” to compensate for the local PABX delays. In the case of cell phones, the number entered must be a service providers SMS (Short Message Service) number.

- The driver will *re-organise* the phone-list at run-time so that the last working number is always on top. This works well if 3 or more numbers are provided, the driver will try a number and then demotes it to the bottom of the list if the number fails to result in a successful transmission. The Agent Server must be running in an account with Administrator rights for the number-list to be auto re-organised in this way.

- SMS Dial-up number for MTN : The SMS number is 0839009000, only specify the cell number in the user code, omit the 083 prefix.

- If using the new MTN 0839009002 number to send to 073 numbers; specify the numbers 0831234567 and 0731234567 in the notify agent as 831234567 and 731234567 respectively.

- SMS Dial-up number for VODACOM : The SMS number is 0822442244, only specify the cell number in the user code, omit the 082 prefix.

- AUTOPAGE : The number depends on area you are in.

How to upgrade from the old pager driver.

1. Take a note of the driver settings for all devices configured on pager, and then delete all the devices in Driver Setup (not in the Agent Server configurator!).

2. Add the devices back, but this time using the 'User Defined' driver, instead of pager. Select your protocol from the list in the driver setup and click on "load protocol defaults".

3. Now change any settings as necessary, click on "Modem" and edit the dial-up number if needed. "OK" the changes and a message will appear indicating that performance counters were successfully installed (first time only).

4. If any errors occur in this procedure, log off and log on as administrator and try again.

5. Run your agent server again as normal, if you are running as a service, we recommend stopping the service via control panel and running the server as a window from the icon in your start menu until it is working correctly.

6. Verify correct operation.

7. You can now delete the old pager.dll file, or rename it to have the .BAK extension.

Wiring

For standard modems, use the modem cable supplied with the modem, for other devices see manufacturer documentation. The normal modem cable provides pin-to-pin matching, and is often also referred to as a null modem cable.

Supported Addresses

The user driver does not have a strict addressing mechanism, addressing individual I/O is only required when the driver is used in a text protocol to acquire data.

For pager and SMS, use scan addresses as Notify : *n* , where ***n*** is a number from 1 to 99999.

When acquiring Digital, String or Analogue values, scan addresses are
Digital : n or
String : *n* or
Analog : *n*
When adding items to the scan point list, there is no difference between using a Notifier Agent and a String Agent.

For sending user strings to the script to use in building messages:

User: n, where 1<=n<=10;

These must be string slots that are used.

Take care when using these values that they are scan inhibited as the driver will do an empty string update at device startup to trigger an output initialize if enabled.

Agent, SMS alarm and Scanning Configuration

Using the Notifier Agent

A very useful feature of the UserDrv driver is its ability to route Adroit information (alarms, values, statuses, messages etc.) to the driver which sends them as an SMS. Adroit makes use of the Alarm and Notifier Agents to do this. The format of the text message that the Notifier Agent sends to the driver is configurable. It can be a simple analogue value, a string, or, as is most useful, actual Adroit Alarm and event messages.

This section explains how to set up a simple test of the alarming and format so that the message text can be passed on to the driver for transmission, it will then cover how to use this in a process alarming or reporting strategy.

- Creating the agents

    1. Create a new Analogue agent called TESTANALOG. Edit the Analogue Tag and configure Low and High alarm levels at 40 and 60.

    2. Create a new AlarmList agent called MYALARMLIST.

    3. Create a new Alarm agent called MYALARMAGENT by selecting the defaultAlarmAgent and using the “Copy” button.

    4. Create a new Notifier agent called MYNOTIFY. Edit the Notifier agent and put the cell-phone number in the edit box entitled “User Code”. This code must be 7 numbers with no spaces dots or dashes. Check the box entitled “Transmit an Alarm when it is raised”. We shall assume the default alarm information is to be transmitted in the text message in this test.

- Configure an alarm

    1. Edit the Alarm agent MYALARMAGENT. You will automatically be in the “Routes” tab, add and select route 2. Under Output Destinations, go to the edit box entitled “Alarm Lists”. Select the AlarmList agent MYALARMLIST.

    2. Go to the edit box entitled “Output”. Select the Notifier agent MYNOTIFY.

    3. Confirm your changes and move to the Analog tag TESTANALOG.

    4. Click on the configurator “Alarm” button to open the alarming dialog. Select the alarm agent MYALARMAGENT. In the “Agent Type” drop-down box, select the Analogue tag type. Add Low and High to the current alarms. Ensure that route 2 is selected.

- Scanning the tag

    1. Finding the notify agent MYNOTIFYin the Configurator, click on “Headder”, turn output-enable on.

    2. Then click scan and scan it from the device we have configured. (If you see a message “Device does not exist create automatically” , select “Yes”.) Use the scan address “Notify : 1” (no quotes) and select the rawValue slot, make the scan-interval zero. Click on “Scan” to add the I/O point.

    3. Click on the “Details” button and verify that the device is started, if not start it now. Accept these settings as needed.

- Testing the driver operation

    1. Open the driver datascope, to do so you can either run the driver monitor launcher and select USERDRV from the list there to open a monitor, or use the “Diagnostics” button in the Adroit Setup program.

    2. In the datascope program, select option 1 which is for datascope, the communications analyser screen will appear. If it does not, the driver could not open the communications port or the driver could not be found, troubleshoot accordingly. If the window TITLE displays “HEX datascope …” then hit ‘A’ on the keyboard to force it to toggle modes into ASCII.

    3. Leave this window running. Go back to the Configurator window, and edit the notify agent. Click on the ”Advanced” button and type in “Hello world” in the format field (if there is already text in the field, add a “dot” period” to the field each time to cause a change).

    4. Click on the Test button (subsequent test messages must differ), and then Alt-Tab back to the Datascope (communications analyser) window. A message must appear indicating that the driver will attempt to start sending. If not go to troubleshooting. As relevant, the modem will start to dial and connect to the gateway or terminal server, watch the display as it does this to determine where if anywhere it goes wrong.

    5. If the SMS does get sent this part is working, else verify the modem operation in some other way.

- Testing the Alarming setup

    1. Edit your alarm agent, and place a value of 100 into it, thereby sending it into high and high-high alarm, click “Update”, then change the value to zero again to generate low alarms.

    2. If this does not result in an SMS being sent, create a new mimic, and display a text value associated to MYNOTIFY.rawValue on it, the last alarm text should thus be displayed, if your “Hello world” is still in there, re-check the alarm configuration steps. The actual text shown should be the subscriber user/cell code, a comma ‘,’ and the alarm text, which is usually preceded by the time.

The Notifier Agent can also support User entered messages. Simply type the message into the value slot from an Operator Action, “Set value” and the message will be sent when clicking on the OK button. Note, to re-send an old message, you must change the message content by e.g. adding a space at the end of it.

Message limitations

Limitations may apply to the content of messages, the driver will NOT convert illegal characters in a message. Generally only use keyboard characters. The message length is limited to 79 characters, including the device code and a comma added for internal purposes.

Protocol / Type of device	Illegal characters for the service
AutoPage	‘<’ , ‘>’ and ‘:’
GSE / Golay	<SPACE>, the space character

GSM, MTN and Vodacom	'[', ']' and '{', '}' and some control codes etc depending on the receiver's phone software.

You may not use any of the above codes in the message to send, the "User Code" field or device agent name. Use the special variable <REPLACEMENTCHARACTERS> to work around these limitations. (See section: ***Custom Protocols***.)

Tag Graying

Scanned strings will not be greyed in the normal fashion used to show communications failures. Specify a non-zero scan-time to ensure that tags are always shown as enabled, even after a failure the tag will stay healthy. See the table of script commands for other ways to stop graying from occurring.

Data Acquisition Devices

This section covers devices such as scales and indicators using ASCII protocols. The driver works in much the same way with the scripts, only it allows data to be acquired in the script itself. Data cannot be sent via this mechanism. Only BOOLEAN and REAL slot types can be read in, this means Analogues and Digitals. Typically the following driver settings are used.

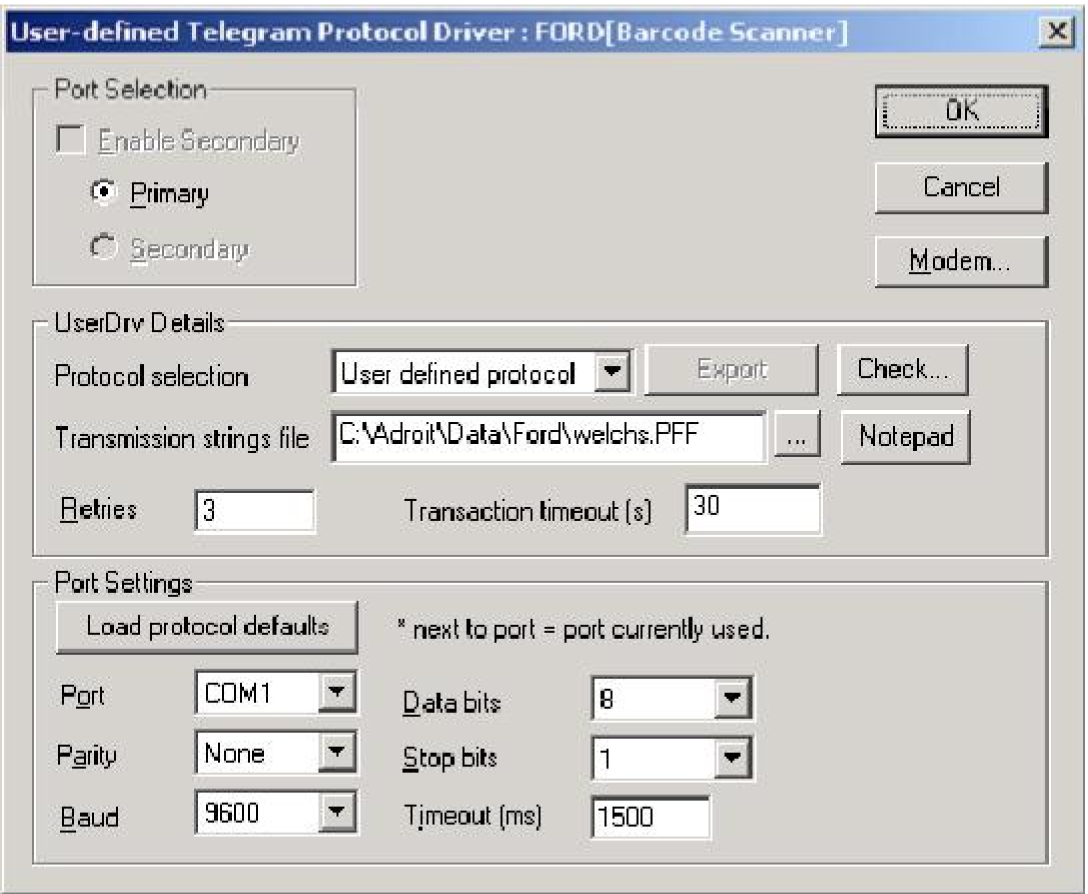


Fig 3

The sample script for the Massamatic scale unit is shown below:

```
REM - Protocol definition for mass scale
<PROTDESC>=MASSAMATIC Scale
REM DO NOT MODIFY THIS FILE!!
REM The protocol is as follows:
REM 01 STX =
REM 02 sign = +/-
REM 03-08   6 digit value with no decimals
REM 09 decimals = number of decimals or space
REM 10 constant k
REM 11 constant g
REM 12 over-range/motion = * or ' '
REM 13-16 CRC = sum of characters 2-12, add 0x30 to each digit
REM 17 END = Carriage return 0x0D
REM 18-22 other data not needed
REM
<UNSOLICITED>=Y
<DATAPRESENT>=Y
<CR>=0x0D
<STX>=0x02
<DELAY>=1000
<RX1>=<STX><PROTOCOLVALUE1>kg<PROTOCOLVALUE2><CR>
<TX2>=<LASTSTEP>
<PROTOCOLVALUE1>=<PROTOCOLSIGN><PROTOCOLANALOGUE>6<PROTOCOLDEC
IMALS><ENDVALUE>
<PROTOCOLVALUE2>=<PROTOCOLDIGITAL><ENDVALUE>
```

This script is parsed starting from whatever character is at the start of the message (<STX> in this case), the valid telegram is assumed to start at that point. An extra <STX> elsewhere in the message will result in incorrect interpretation. The driver then parses values as they appear in the RX string received, the sample above has been simplified by using 2 variables to show the 2 values (PROTOCOLVALUE1, PROTOCOLVALUE2) acquired more clearly. See the table of commands and variables below for detail on other aspects of this script.

Sample for ADAM 4000 modules

```
REM - Protocol definition for ADAM 4000 modules
<PROTDESC>=ADAM 4000 module
REM DO NOT MODIFY THIS FILE!!
REM The protocol is as follows:
REM Request:
REM #AAN
REM    the AA is the module address.
REM    the N is optional, if not provided, the module responds with all of its channels
REM If 'N' is provided, the module responds with only the requested channel.
REM The response format is:
REM >+123.456<CR>
REM >+123.456+789.012+345.678+901.234<CR>
REM >+123.45644444+789.012+345.678+901.234<CR>
REM >+1.5011+0.9529+0.5507+0.2206+0.3280+0.1253+0.0501+0.0209<0D>
<VARDEFAULTPARITY>=N
<UNSOLICITED>=N
<DATAPRESENT>=Y
<NUMERICDELIMITERS>=+-<CR>
<CR>=0x0D
<STX>=#
<RESP>=>
<DELAY>=1000
<MODULE1ADDR>=01
REM Using all channels at once
<TX1>=<STX><MODULE1ADDR><CR>
```

```
<RX1>=<RESP><PROTOCOL1><PROTOCOL2><PROTOCOL3><PROTOCOL4><PROTO
COL5><PROTOCOL6><PROTOCOL7><PROTOCOL8><CR>
<TX2>=<LASTSTEP>
REM Request each channel individually.
REM <TX1>=<STX><MODULE1ADDR>6<CR>
REM <RX1>=<RESP><PROTOCOL1><CR>
REM <TX2>=<STX><MODULE1ADDR>1<CR>
REM <RX2>=<RESP><PROTOCOL2><CR>
REM <TX3>=<STX><MODULE1ADDR>2<CR>
REM <RX3>=<RESP><PROTOCOL3><CR>
REM <TX4>=<STX><MODULE1ADDR>3<CR>
REM <RX4>=<RESP><PROTOCOL4><CR>
REM <TX5>=<STX><MODULE1ADDR>4<CR>
REM <RX5>=<RESP><PROTOCOL5><CR>
REM <TX6>=<STX><MODULE1ADDR>5<CR>
REM <RX6>=<RESP><PROTOCOL6><CR>
REM <TX7>=<STX><MODULE1ADDR>6<CR>
REM <RX7>=<RESP><PROTOCOL7><CR>
REM <TX8>=<STX><MODULE1ADDR>7<CR>
REM <RX8>=<RESP><PROTOCOL8><CR>
REM <TX9>=<LASTSTEP>
REM the variables formats
<PROTOCOL1>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL2>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL3>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL4>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL5>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL6>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL7>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
<PROTOCOL8>=<PROTOCOLSIGN><PROTOCOLFLOAT><NUMERICDELIMITERS><EN
DVALUE>
```

Typing Scan-Addresses

To get a tag to update from this data, run the driver with your script and then look in Datascope where the driver reports the values it got out of the received string, Look for the statement “type=*X*”.

The driver handles 3 data types "Strings", "Digitals" and "Analogs" in that order the type numbers are 2,3 and 4 respectively. To get the data to update adroit correctly you need to match the type in your scan-address with the type that the driver detects the data as.

Hence: If the driver picks up "analog" or "4" you need to type "Analog:1" not "Notify:1" as a scan-address. If the driver picks up "digital" or "3" you need to type "Digital:1". (All related to what types appear in the script.)

Lastly the "index=x", will tell you what number to put in after the colon ":".

Cellular GSM Modems

Introduction to GSM

After extensive use as a custom protocol, the script for GSM or cell modems is now supported as built-in. Note that when using a cell modem, you do not enable the “use modem” tick-box. The message is actually sent using special ‘AT’-like codes that access the SMS functionality of the GSM. Hence you only pay for the SMS, and no dialing actually occurs, although this mechanism still remains available since the GEM modem is a normal modem too.

Configuring the GSM modem

Some effort is required to set up the modem for initial operation; this is a once-off effort. Please note that this procedure may become irrelevant in future.

Although the driver has an SMS/Cel-modem protocol selection, this "script" for the driver does not pre-configure the modem, in order to save time on each transmission. The necessary steps to perform are laid out here, along with the meaning, once completed, even turning off the device will not loose these changes.

The necessary steps to perform are laid out here, along with the meaning, once completed. Even turning off the device will not loose these changes. Actual HyperTerminal instructions may very depending on which Microsoft operating system you use. Exit all un-necessary applications before starting.

If not installed, install HyperTerminal, as follows:

From control panel, select Add/remove programs.

Click on the "windows" or "components" tab, and select HyperTerminal under "communications" to install it.

Select windows start/programs/communications/HyperTerminal, HyperTerminal will launch.

Type in a connection name (you can save this profile if you need to use it again) If this is the first time you run HyperTerminal, it may ask to install/configure a modem, ignore this step by providing any needed information or skipping as needed. You do not want to be installing a modem driver at this point.

Use default settings at all these steps, only select the comm. port you are using. Baud rate is not important, select 9600 if in doubt; no handshaking, 8 data bits, 1 stop and no parity. When you see "0" in this explanation it is a zero, not uppercase "o", case is not important in these commands, but uppercase is used for simplicity. If them instruction says to turn the modem off, do so for at least 5 seconds.

1. Click on the "Connect" toolbar icon; then type "ATZ" (no quotes) followed by Carriage-return <ENTER>
   You should see the text ATZ as you type it, and the next line should read OK.
   If you see your typing ATZ appearing twice, click on the menu "File/properties" click "ASCII setup", and remove the check from the box "echo typed characters locally". OK the settings, and click the "connect" toolbar icon.
2. If the modem does not respond:
   Disconnect HyperTerminal and select 2400 baud, reconnect. Switch the modem off, then on, and type AT<ENTER> in (the AT MUST be upper-case here) 3 times over. Doing so allows the modem to train to a new baud-rate, which it will loose if re-initialized.
   Type AT+IPR=9600;&W<ENTER>, then switch the modem off and on again, change the HyperTerminal baud-rate back to 9600, it should respond properly on the first command now. If the modem still returns garbage, type AT+ICF=3,4<ENTER> to set up 8 data-bits 1 stop and no parity. Save this using the W command as above, and cycle power on the modem.

3. Type ATE1 and <ENTER>, the unit will respond with OK. Now we can build on this, type AT&D2<ENTER> this turns the handshaking to the default we want.
4. Then AT+CMEE=1<ENTER> this turns error reporting mode , (which the datascope displays as a number if an error is reported) , the unit will respond with OK.
5. Then AT+CMGF=1<ENTER>, this puts the modem into Text mode. the unit will respond with OK.
6. Then type ATE0&W0<ENTER>, this will turn echo off, and save the modem settings to profile 0, also type ATE0&W1<ENTER> to save into profile 1 (this is because some modems have profile 0 and 1, and others have 1 and 2). If using a Falcom modem, the “W” must not be followed by a 0 or 1, the modem only has 1 profile.
7. Turn the modem off and on.
8. In HyperTerminal, type AT+CMGF?<ENTER>, the unit will respond with AT+CMGF: 1, this verifies that our settings have been saved, else go back to step 5.
9. Finally to test that all is working.
10. In HyperTerminal, type AT+CMGS=”0821234567”<ENTER> , use your own mobile number here, it must appear between double-quotes. The unit will then prompt with a “>” (greater-than) sign.
11. Type in a test message, for Example, “This is a test message”, do not use extended characters or escape codes. (You may use capitals though.) Finnish off your message with the <CTRL-Z> key combination. To do this, hold down the <CTRL> key, and tap ‘Z’. This adds character x1A (26 dec), you do not need to hit <ENTER>. The unit should respond with a “CMGS:” and “OK” or else an error code. If you get an error code back, the message center has probably not been set up.

12. Your mobile will receive the message, normally within 5 seconds.
13. If not, you want to verify signal strength and check your antenna.
    Type AT+CSQ<ENTER> the modem will respond “OK + CSQ:21”. The value must lie between 11 and 31, the higher the better.
14. Close HyperTerminal. Run your Adroit Agent Server, and send a test message.
15. If all else fails Issue AT&F<ENTER> to reset the modem dialer back to factory defaults. And follow all the steps again.

NOTES:

If the CMGF command gets a response with error 311, enter the pin code for the unit. This is best done using a phone, and then turning off the SIM checking feature in the security menu, this cannot be disabled via HyperTerminal on most GSMs. For now, type AT+CPIN=xxxxx where xxxxx is your pin, if the ATZ command is sent or the unit turned off or SIM removed, pin must be re-entered.

On a Wavecom and Falcom the at&w0 will not work, use at&w instead.

Custom Protocols

The PFF file

The PFF or Protocol Format File is not a script, it defines variables which are used to control the communication. The driver uses these variables to determine what to do when communicating and how to build the messages to send and receive.

How to write a format file

The format file consists of comments and variables, most of the latter are pre-defined. The author may also define new variables.

Variables are declared as:
<VARIABLE_NAME>=value
Variable values are case sensitive, names are not.

Telegrams are the basic unit used to describe the communication process. A single transaction usually involve something to transmit and something to be received, these are called telegrams, there may be as many as 10 pairs of these altogether. The driver will process them in sequence (i.e. TX1, RX1, TX2, RX2, TX3, TX4). If a response is received that matches a different one to that expected, but it does match another receive string, the driver will jump to the relevant step that follows it. Thus the string for each RX string needs to be unique in some way, the RX strings do also not have to be complete, and a sub-string may be used.
E.g. <RX1>=Logon:
Would suffice if a prompt such as “Enter user name to Logon:” was expected from the remote device so long as the word “Logon:” was unique and would not appear elsewhere in other communications steps.

The step consists of a TX and RX part, which have the same step number. Not all steps need to be present, you can leave a step blank or remove it completely (missing steps are still run, they are simply run through quickly). The TX telegram is transmitted after it has been parsed, then the RX telegram is parsed then checked or matched against the received data. Only telegrams can consist of variables by imbedding them, you may only imbed variables in a telegram, not in other variables. Variables within variables will not be parsed or replaced with their corresponding values.

E.g. <MYVAR>=Joe soap
<TX1>=<CR>:<MYVAR><CR>
That would be translated into "↵:Joe soap↵"

Telegrams are declared as:
<TXn>=value

Where 'n' is a telegram number and RXn may be used for RX telegrams a special variable <LASTSTEP> is used to denote the last telegram. A maximum of 100 telegrams may be defined, this is the same as 50 steps.

Single character values coded as 0xNN will be converted to the hex equivalent. To place a hex code at any position other than 1 in a telegram/step, create a new variable and use that instead.

CRC values are calculated using an algorithm, there is currently only one defined <CRC1> - it is the Telstra Mobile TAP Protocol CRC algorithm. No other ones have been defined yet.

Some values take the form of commands:
<GOTOSTEP> specifies a step to jump to. (The driver will jump to the TX step of the same number.)
<ERROR> indicates that transaction failed. (The driver will fail the transaction.)
<DEBUGMESSAGEn> tells the driver to display debug message ‘n’ before processing this step.

If the remote modem hangs up on the last RX step (I.e. the remote end responds and then hangs up immediately) then the next Step TX must be <TXn>=<LASTSTEP> because the driver will otherwise check for carrier and fail.

Predefined values

The following is a list of predefined values. Do not modify any pre-defined value that is marked as “assigned at runtime”. Some values perform special functions as well. (R=runtime, V=Variable, C=Command or instruction)

Variable	Meaning or description	Code
<DEVICE>	Assigned from device name at runtime.	R
<MSG>	Message assigned at runtime.	R
<CODE>	User code/number/id assigned at runtime.	R
<PASS>	Assigned at runtime. This variable may be used as a password	R
<DELAY>	A delay or wait that is inserted (default=0) wherever this statement is encountered in a RX or TX step, the driver will wait for the specified delay, this variable’s value can be set as <DELAY>=1000 for example to produce 1 second delay on a RX by using: <RX1>=<DELAY>Reply	C/V

<STX>	Start transmission character from file (default = x02).	R/V
<ETX>	End of Transmission character from file (default = x03).	R/V
<EOT>	End Of Text character from file (default = x04).	R/V
<ACK>	ACKnowledge character from file (default = x06).	R/V
<NAK>	Negative AKcnowledge character from file (default = x15).	R/V
<ESC>	ESCape character from file (default = x1B).	R/V
<CR>	Carriage return character (default = x0D).	R/V
<LF>	Feed character (default = x0A).	R/V
<LASTSTEP>	The text "LASTCOMMSTEP". Place this on the last step. E.g.: <RX3>=<LASTSTEP>	C
<PROTDESC>	The text "No Description". E.g.: <PROTDESC>=My Custom Protocol.	V
<CONNECTDELAY>	Delay to wait after connecting and a carrier is achieved (default = 4000 milliseconds).	V
<CRCn>	CRC routine “n” calculated by the driver (assigned at runtime).	C
<TXn>	Step n transmit definition	V
<RXn>	Step n receive definition	V
<ERROR>	Indicates that this telegram is an error, end transaction immediately. Add it to the end of a line.	C
<GOTOSTEPn>	Step number to go to on error, usually 1 to retry. Not the transaction will still end if the total transaction time expires, if the “gotostep” is equal to the last step, the transaction ends in error. Add it to the end of a line. E.g.: <GOTOEND>=6 … <TX3>=Hello World<GOTOEND> … Notice that the variable is assigned a value, and then gets used in a step.	C
<DEBUGMESSAGEn>	A debug message number to display before executing this step. E.g.: <DEBUGMESSAGE1>=My Step1 … <TX3>=Hello World<DEBUGMESSSAGE1> Notice that the variable is assigned a value, and then gets used in a step. The message text is also	C/V

	event logged.	
<VARDEFAULTNUMBER>	**Load-able default:* String defining the default number to dial, if this is anything other than “0,” then the “use modem” option is also enabled at the same time.	V
<VARDEFAULTBAUD>	**Load-able default:* Default baud-rate. (defaults to 9600)	V
<VARDEFAULTPARITY>	**Load-able default:* Default parity setting, must be “E”, “N” or “O”. (defaults to “E” if neither)	V
<VARDEFAULTSTOP>	**Load-able default:* Default stop-bits: 1 or 2. (defaults to 1)	V
<VARDEFAULTDATA>	**Load-able default:* Default data bits: 7 or 8. (defaults to 8)	V
<VARDEFAULTTIMEOUT>	**Load-able default:* Default communications timeout (ms). (defaults to 2500 milliseconds)	V
<VARDEFAULTCONNECT>	**Load-able default:* Default connecting timeout (ms). (defaults to 30000 milliseconds)	V
<VARDEFAULTTRANS>	**Load-able default:* Default transaction time (s). (defaults to 30 seconds)	V
<VARDEFAULTMODEMSTR>	**Load-able default:* Default modem initialisation string (if empty, it has no effect)	V
<VARDEFAULTIDLE>	**Load-able default:* Default idle time to wait before hanging up (ms). (defaults to 30000 milliseconds)	V
<UPPERCASE>	Converts this entire telegram to upper case. For transmit messages, the message is converted before it is transmitted. For receive messages, the message compare mask is converted to Upper Case, but not the received data. E.g.: <TX1>=Convert this to Upper Case!<UPPERCASE> Reads: CONVERT THIS TO UPPER CASE!	C
<REPLACEMENTCHARACTERS>	String containing illegal chars and their legal replacements. Example use “:=&+” to convert all ‘:’ to ‘=’ and all ‘&’ to ‘+’. May not include any variables, and only applies to the message field in Adroit, not the user code.	V
<UNSOLICITED>	Default = ’N’. If ‘Y’ then the driver will wait for in-coming data else it will poll.	V
<DATAPRESENT>	Default = ’N’. If ‘Y’ then the driver will poll or wait for data. Used together with <UNSOLICETED> this flag enables the use of Analogue and Digital I/O acquisition.	V
<DECODE RESPONSE>	Default=’N’. If ‘Y’ then the driver will not poll or wait	V

	for data. It will however interpret the response to a transmission as data. Used together with <UNSOLICITED> and <DATAPRESENT> to extract data from responses to transmission. Only available for TCP/IP mode devices	
<PROTOCOLSIGN>	A value in the telegram must correspond to '-', a '+' or a space to indicate the sign of a value.	R
<PROTOCOLANALOGUE>	A value in the telegram must correspond to '0' through '9' to indicate a value. Must be followed by a width specifier, e.g. '<PROTOCOLANALOGUE>8' would mean 8 character analogue.	R
<PROTOCOLFLOAT>	A value in the telegram must correspond to '0' through '9' and may contain a '.' At the decimal position for a value. Must be followed by a width specifier, e.g. '<PROTOCOLFLOAT>8' would mean 8 character analogue.	R
<PROTOCOLDECIMALS>	A value in the telegram must correspond to '0'thru '9' or space ' ' to indicate the multiplier.	R
<PROTOCOLDIGITAL>	A value in the telegram must correspond to '0' or '1' or '*' to indicate a value.	R
<PROTOCOLSTRING>	A value in the telegram will be placed into string tags scanned as "String:X". The variable definition must contain <NUMERICDELIMITERS> that will be used as string delimiters, or must have a length immeditaely following it for fixed-length strings.	R
<ENDVALUE>	This indicates to the driver to use the value information it has received (e.g. sign * analogue value * decimals^10) and update the server. Each consecutive <end> increments the scan address index (number at the end of the scan address) that will be updated. The values are sent in the order they are parsed from the received data.	C
<TRANSMITTYPEDDELAY>	Default = 0. A delay in milliseconds to be imposed between each character sent to the device. This delay simulates the action of manually typing the string. A typical value would be 50 milliseconds.	V
<TRANSMISSIONLOGFILE>	Default is Empty, if supplied, all controls or outputs get logged to the filename specified. E.g. <TRANSMISSIONLOGFILE>=SMSLOG.TXT	V
<COMPRESSSPACES>	Driver will remove all spaces before processing numeric value.	R
<TEXTTERMINATION>	"End-Of-text" code to use when the data has no TX step and arrives at high speed continuously.	V

<BARKBADAFTERTO>	Default value N, this variable can be set to Y to stop tags being greyed after the driver times out and runs out of retries.	V
<PURGEAFTERRX>	Allowable values are 0 and 1, if not 0 (the default), the driver will purge all remaining characters in the buffer once it has the end-of-text marker specified in **ENDVALUE**	V
<REVERSEANALOGS>	Allowable values are 0 and 1, if not 0 (the default), all analogue values are swapped around before processing, this can be used to convert right-left printed values.	V
<USER VAR 1>	User variable 1. Used to build strings. Writable from the Agentserver	V
<USER VAR 2>	User variable 2. Used to build strings. Writable from the Agentserver	V
<USER VAR 3>	User variable 3. Used to build strings. Writable from the Agentserver	V
<USER VAR 4>	User variable 4. Used to build strings. Writable from the Agentserver	V
<USER VAR 5>	User variable 5. Used to build strings. Writable from the Agentserver	V
<USER VAR 6>	User variable 6. Used to build strings. Writable from the Agentserver	V
<USER VAR 7>	User variable 7. Used to build strings. Writable from the Agentserver	V
<USER VAR 8>	User variable 8. Used to build strings. Writable from the Agentserver	V
<USER VAR 9>	User variable 9. Used to build strings. Writable from the Agentserver	V
<USER VAR 10>	User variable 10. Used to build strings. Writable from the Agentserver	V

- Load-able default values are loaded into the driver configuration when you click on the “Load protocol defaults” button. For example entering the baud-rate into the PFF file and instructing users to simply click on the “load defaults” button will make using a custom script easier and less error prone.

- The script variables UNSOLICITED, DATAPRESENT, PROTOCOLSIGN, PROTOCOLANALOGUE, PROTOCOLFLOAT, PROTOCOLDECIMALS, PROTOCOLDIGITAL and ENDVALUE are only used in polling/telemetry acquisition driver scripts.

Examples:

Example of the use of *TEXTTERMINATION* .
<TEXTTERMINATION>=<CR>
<RX1>=<STX><PROTOCOLVALUE1> kg <PROTOCOLVALUE2><TEXTTERMINATION>

Barcode scanners

Because a barcode ultimately consists of a sequence of numbers, some of which may be leading zeroes, and even may exceed the number of mantissas stored for floating-point numbers, strings must be used.

From revision 22 onwards of the driver, strings are also supported as data input values, they must be scanned as “String:1” not as “Notify:1” to work correctly. Below is the PFF script supplied to read the barcodes scanned by a handheld scanner. The driver actually times out when the scanner is not sending a code, but this does not affect the value, since no error occurs, and the tag will not “gray” out. Use a zero scan interval when adding the scan-point, using any string slot tag in Adroit. A triggered Script can be used to process the resulting barcode.

```
REM - Protocol definition for Welch-Allyn barcode scanner
REM string tag version
<PROTDESC>=Barcode Scanner
<UNSOLICITED>=Y
<DATAPRESENT>=Y
<STX>=0x02
REM<STX>=Y
<CR>=0x0D
<LF>=0x0A
<NUMERICDELIMITERS>=+-<CR>
<DELAY>=1000
<RX1>=<STX><PROTOCOLVALUE1><CR><LF>
<TX2>=<LASTSTEP>
<PROTOCOLVALUE1>=<PROTOCOLSTRING><ENDVALUE>
```

Limitations

- If the received message contains any NULL (zero) characters, these are replaced with a (tilde) ‘~’ before processing.
- No defined variables except for TX and RX steps may contain imbedded variables.
- Hex coded values are only allowed as the 1st character of a variable.
- If consecutive TX and RX steps are empty, the script ends at that step. I.e. TX4 and RX4 are empty or not there, the driver assumes that this is the last step. If RX3 and TX4 are empty, but there is something defined for RX4, then the script will continue.
- Limited to 99 steps.
- Maximum length of a definition in the script is 1024 characters long.
- Maximum of 99 variable definitions

Message protocols

Datascope messages are displayed in ASCII. These messages are meaningful descriptions of the paging step currently engaged and are not a reproduction of the protocol. The messages displayed are particular to the paging protocol selected.

Zetron protocol

An example of a session using the Zetron Encoder is given below:

TX	:	<CR>
RX	:	ID=
TX	:	PG80000
RX	:	<ACK><CR>
RX	:	<ESC>[p<CR>
TX	:	<STX>str1<CR>str2<CR><ETX>str3<CR>
RX	:	<ACK><CR>
TX	:	<ESC><CR><EOT>* bye *<LF><CR>

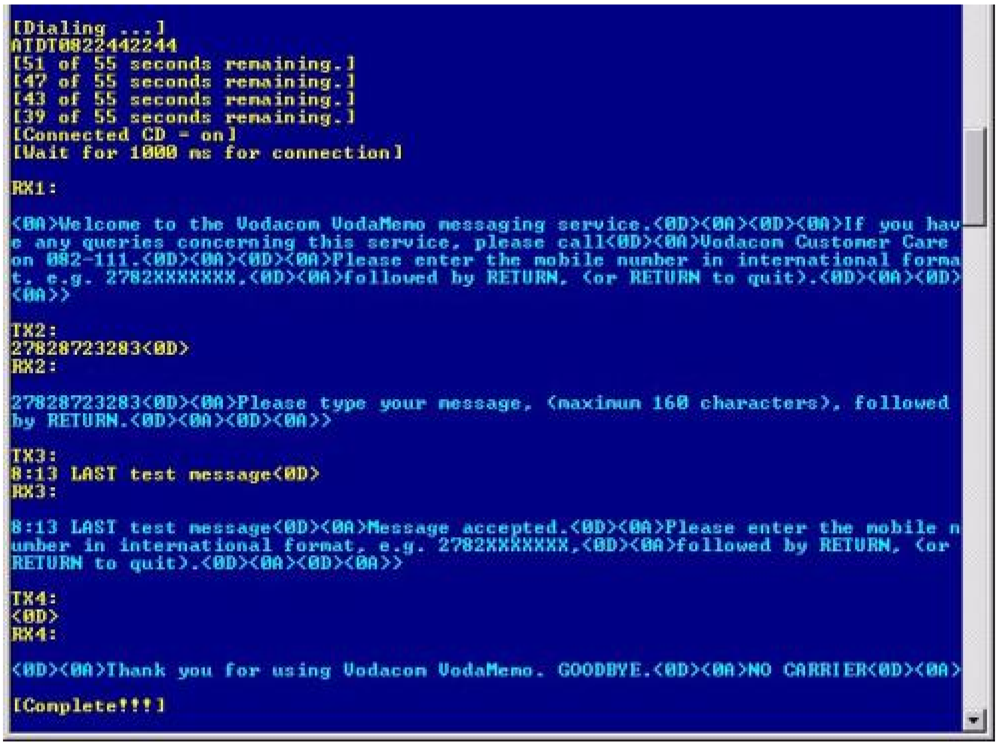
Where str1 = Pager Field
str2 = Actual ASCII text message
str3 = Checksum

AutoPage protocol (*The old protocol)

TX	:	X.28<CR><CR>
RX	:	NUI?
TX	:	90102043YAC 11631309
RX	:	CONNECTED TO DTE AT 11631309
TX	:	<PL:aaa:pppppp:tttttttt:cccccccc:3:1::7:message><CR>
		a=id ; p = password ; t = tag(device) ; c = code
RX	:	<OK:aaa:tttttttt:cccccccc:00: :OK><CR>

- The communications settings and Speed-link (X.28) portions of this procedure have been changed.

Vodacom protocol

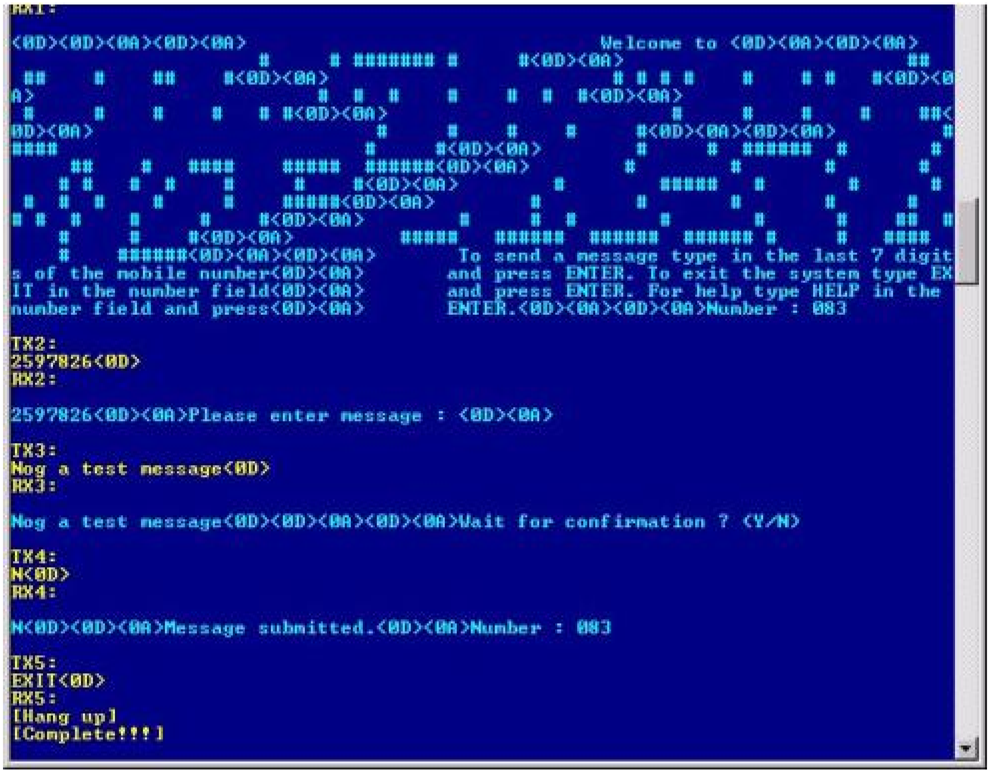


The Vodacom protocol must be modified for SMS'ing different number prefixes, 082/072 etc.

1. Use the export button in the driver setup to save the protocol to a file.
2. Edit the file in notepad, REM the relevant lines out, and un-comment the relevant lines.
3. Save the file, and ensure that it gets saved without a .TXT extension
4. In the driver setup again, select the protocol as "custom/user-defined", and browse in the file you just saved in the edit-field for this purpose which will have become active now.
5. Start/re-start the agent server.

You will need to create one device agent and setup in this fashion for every prefix you need to dial. Alternately, edit the file, and remove the prefix completely, include the prefix in the notify agent used instead. This approach however reduces by 2 characters, the SMS length that can be sent.

MTN protocol



The MTN protocol can not be modified to SMS mobile numbers other than those with an 083 prefix

Modem reference

Abbreviated modem reference: please refer to the user manual that accompanies the modem, since details may vary. Commands are prefixed with AT and followed by a carriage return.

Abbreviated reference based on Rockwell chipset modems

En		Sets local echo.
	E0	Echo OFF
	E1	Modem displays keyboard commands (the default)
Dn		Dials the specified phone number. Includes the following:
	L	Dials the last dialed number.
	P	Pulse (rotary) dial
	T	Tone dial
	,	(Comma) Pause,
Ln		Controls speaker volume (for internal models only).
	L0	Low volume
	L1	Low volume
	L2	Medium volume (the default)
	L3	High volumer
Mn		Operates speaker.
	M0	Speaker always OFF.
	M1	Speaker ON until CONNECT. (the default)

	M2	Speaker always ON.
	M3	Speaker ON after dial, until CONNECT.
Sr=n		Sets register r to n.
Sr?		Displays contents of S-Register r.
S$		Displays a list of the S-Registers.
Z		Resets modem.

Additional ampersand & commands

&Cn		Controls Carrier Detect (CD) signal.
	&C0	CD override
	&C1	Normal CD operations (the default)
&Dn		Controls Data Terminal Ready (DTR) operations.
	&D0	DTR override (the default)
	&D1	DTR toggle causes online Command mode
	&D2	Normal DTR operations
	&D3	Resets on receipt of DTR
&Wn		Writes current configuration to NVRAM templates.
	&W0	Modifies the NVRAM 0 template (Y0)
	&W1	Modifies the NVRAM 1 template (Y1)

Important S (setting) registers

S0	0	Sets the number of rings on which to answer in Auto Answer Mode. When set to 0, Auto Answer is disabled.
S7	60	Sets the number of seconds the modem waits for a carrier. May be set for much longer duration if, for example, the modem is originating an international connection.
S10	14	Sets the duration, in tenths of a second, that the modem waits to hang up after loss of carrier. This guard time allows the modem to distinguish between a line disturbance from a true disconnect (hang up) by the remote modem.
S11	70	Sets the duration and spacing, in milliseconds, for tone dialing.
S15	0	Bit-mapped register setup. To set the register, see instructions for S13.

Bit	Value	Result
0	1	Disable ARQ/MNP for V.22.
1	2	Disable ARQ/MNP for V.22bis.
2	4	Disable ARQ/MNP V.32/V.32bis/V.32terbo.
3	8	Disable MNP handshake.
4	16	Disable MNP level 4.
5	32	Disable MNP level 3.
6	64	MNP incompatibility.
7	128	Disable V.42 operation.
	136	Disable V.42 detect phase.(The sum of the values of bits 3 and 7.)

Troubleshooting

PFF scripts

Only pre-defined scripts are supported, any scripts browsed in using “Custom Protocol” need to be debugged. The process for debugging a custom PFF file is simple…

1. From Setup, click on “Check…”, a dialog with all the script variables in it appear.

2. Double-click on each user variable labelled “TX1” etc to make sure that the driver can process it correctly. Fix any errors then close this dialog and open it again and re-check your changes.

3. Close the driver setup dialog. If you still have problems continue…

4. Once connected, the driver will wait for a while to allow the server to respond. Then it will commence with the processing of the communication steps. If the driver sends a string that is incorrect or not recognised by the remote device, that device will either not respond or will give an error response. Use the ASCII “DataScope” to determine if this is the case.

5. If problems persist, try exporting all of the built-in scripts and analysing them to find a possible solution. Note: You can edit the script and save it between attempts, since the driver re-processes the script on every transaction.

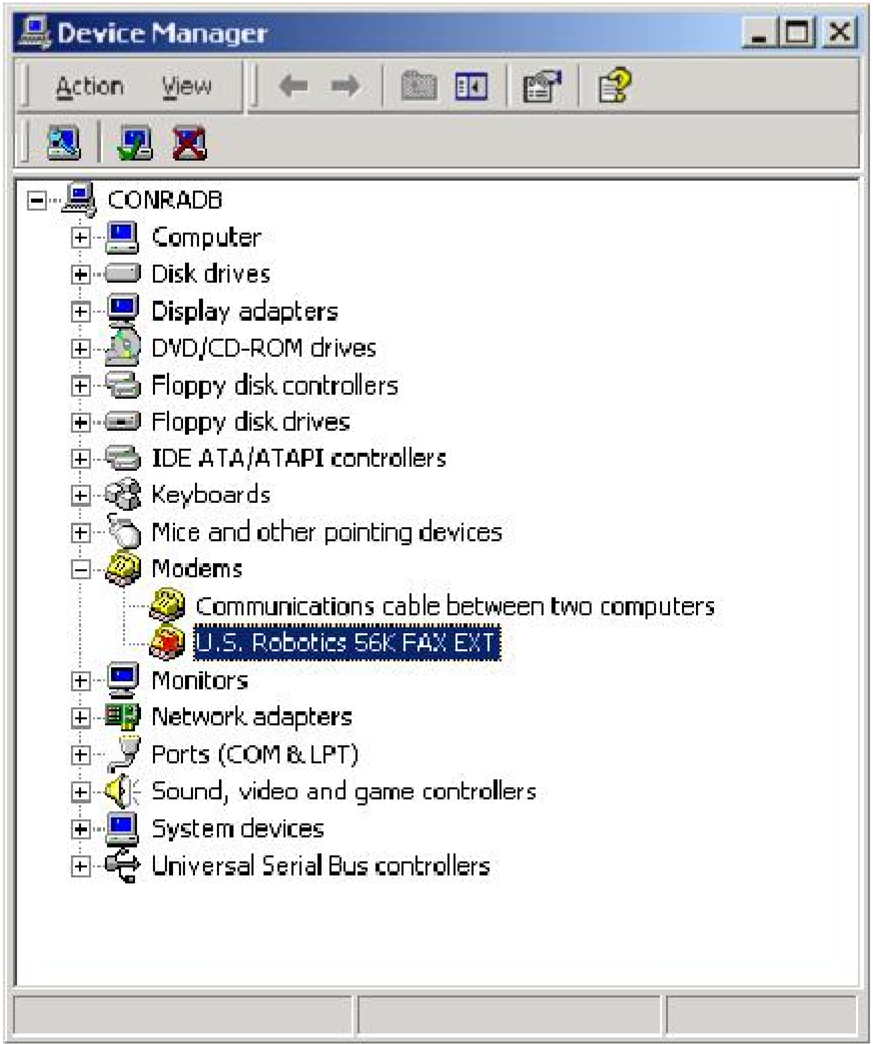
6. Q: My tag does not update, what am I doing wrong? A: If the Datascope is showing a update value and it looks correct, all you need to do is type the scan-address correctly, see the section on Typing scan-addresses above.

7. Q: What do I do if the device’s response for a step is random or un-defined? A: Use an empty RX step, the driver will read and display the response without doing anything else with it.

8. The PFF script is re-loaded every time the datascope displays [Attempt n/n Sending xxxx] or [Acquire I/O data], certain script variables require a re-start to re-load or change.

Modem installation problems

9. Under Windows 2000 the operating system may install a modem driver for you automatically either on start-up or when the modem is plugged in. If the driver dose not see the port, simply disable the modem driver after you have installed it as shown below. Right clicking on the modem and selecting "disable" from the pop-up menu accomplishes this.



General problems

10. Send a short message while the Datascope window for that device is open and in "ASCII" mode, to toggle HEX/ASCII mode hit "A" on the keyboard.

11. If using a modem, make sure that the remote end does answer, if not, try manual dialling of the number from a phone to see if the call gets picked up. You may need to increase the 'connect timeout' value to remedy this.

12. Turn 'ignore dial-tone' on if the dial-tone is not clear or not present. This may also indicate poor line quality and lower connection speeds.

13. Check that you can dial exactly the same number, if dialling "0" to get a line, try using more than one comma after the "0" to allow more time for a line to become available. (1 second per comma)

Multiple messages

14. When sending multiple messages to a pager, if the driver fails to send a message (i.e. the retries get exhausted) and another message is queued within 30 seconds of the failure, the new message is not sent. However any messages already queued before the communication fails will still be sent via scanning to the driver for transmission.

Abbreviations

DT	Dial Tone
DTR	Date Terminal Ready (a hardware signal)
GSM	Global System for Mobile telecommunications
PIN	Personal Identification Number
PUK	Personal Unblocking Key
SMS	Short Message Service

Revision information

Date	Changes
04-Feb-00	Updated the screen-shots and new details.
11-Feb-00	Additional list of commands and variables.
23-Mar-00	Scan-good marking by returning healthy on Reads. Replaceable characters.
6-Jun-00	Scale-driver modifications added.
13-Oct-00	Extra analogue types.
12-Jan-01	Sample ADAM 4000 script.
19-Feb-01	New variable for between-character delays in transmitting of a string.
18-Mar-01	New variable for Text-logging of SMS transmit strings.
5-Feb-02	Cell modem support section added. Driver now includes Cell and Adam scripts.
2 –Mar-02	String support added for Barcode-Scanning. Fixed typo in the Cell-modems section.
4-Apr-02	Added abbreviation section. Added modem disabling if the modem driver gets installs.
25-Jul-02	Added new variable for bad marking
08-Oct-02	Added how to change Vodacom SMS mobile prefix from 082
04-Mar-03	Spelling errors fixed
15-Aug-03	Re-write of simple test and how to set up the Notify.
9-Mar-04	Dialog screenshot updated with new serial port control in rev 31.
17-Dec-04	String reverse, and RX buffer purge variables added in rev 32.
11-Jan-06	GSM modem initialization/pre-configuration section updated.
13-Jun-2012	Added support for TCP/IP and added the <DECODE RESPONSE> and <USER VAR x> identifiers.